# Can We Trust Logged LLM Comparisons? Auditing Selection and Measurement Validity under Model Routing

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

Many AI products choose among several language models for each user request. A router may send routine questions to a cheaper model and difficult questions to a stronger one, leaving a log of the request, the model that answered it, and an automatic score. Teams use these logs to compare models and decide which should handle each kind of request. We argue that such comparisons can fail in two ways, while existing work largely addresses only one. Assignment validity fails when models are judged on different requests. If one model receives harder questions, their average scores are not directly comparable. Measurement validity fails when the score rewards the wrong thing. A grader may prefer longer or more polished answers when the team cares about correctness or user value. Methods for confounded logs use randomized audits to address the first problem, but usually treat the score as valid. Research on evaluator bias studies the second problem, but rarely connects it to model selection decisions from routed logs. We show that a randomized audit isolates routing bias only when the grader makes the same average errors on routed and randomly selected requests. Even then, grader bias remains. We test both failures on two public routing benchmarks. Auditing reduces assignment bias on both, but not measurement bias. Under a prespecified scoring bias stress test, even perfect routing correction selects the wrong model in $14$ to $18\%$ of request types on the two corpora, and in $10$ to $31\%$ across our sensitivity analyses. We also find that corrected scores matter more than the exact audit allocation rule, although the rule matters more as the number of models grows. Finally, the first few randomized measurements can indicate whether a larger audit is worthwhile.

## 1 Introduction

A team runs a product on several language models behind a router, so that cheap models take the routine traffic and expensive ones take the hard traffic [Chen et al., 2023, Ding et al., 2024, Ong et al., 2025]. After a month there is a log with millions of rows, each holding a request, the model that served it, and an automatic score. Someone asks a reasonable question. *For coding requests, which model is best?* The log answers instantly and for free, and the team switches coding traffic accordingly. That answer can be wrong for two reasons that look alike and are not.

**The models sat different exams (assignment validity).** The router decided which model saw which request, and it decided on difficulty. Modern routers make this explicit. They predict per request how well each candidate would do, or how uncertain it is, and escalate accordingly [Feng

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et al., 2025, Zhang et al., 2025, Su et al., 2026, Zheng et al., 2026]. The strong model's average is therefore taken over hard requests and the cheap model's over easy ones, so comparing the averages compares models and exam papers at once. Recent work on routing supervision makes the same observation from the training side, noting that outcomes sampled under a routing policy are a biased view of a model's capability [Lai et al., 2026]. This is the classic confounded-log problem, and randomization is its classic answer.

**The exam measured the wrong thing (measurement validity).** The score in the log came from an automatic evaluator, such as an LLM acting as judge, a string-match grader, or a click-through proxy. Such evaluators have documented systematic preferences, including for their own outputs, for longer answers, and for whichever answer appears first [Wang et al., 2024, Panickssery et al., 2024, Zheng et al., 2023, Chiang et al., 2024]. If the evaluator's ranking differs from the ranking the deployment decision is about, every model sat the same exam and the exam was the wrong one. This is the classic *construct*-validity question [Jacobs and Wallach, 2021, Raji et al., 2021], namely whether the number you measured is the quantity you care about, and randomizing who is graded on what does not touch it.

**What is missing.** Both failures are individually familiar. The methods literature on confounded logs combines an observational log with a small randomized sample and spends an experimental budget where it changes decisions [Athey et al., 2020, Kallus et al., 2018, Rosenman et al., 2018], and the last two years have sharpened both halves, with cross-validated fusion of experimental and observational data [Yang et al., 2025], confounding-robust offline learning that buys online interventions [Li et al., 2025], and decision-oriented active designs [Cha and Lee, 2025, Gao et al., 2025, 2026a,b]. All of it targets the first failure and reports gains against a benchmark score treated as valid. The evaluation literature documents the second, mostly by measuring judges against human labels, without connecting it to a model-selection decision taken from a routed log. Missing is the joint statement a practitioner needs, namely *which* failure an audit repairs, what it leaves behind, and how the leftover shows up in the decision they make. Answering that requires switching one failure on and off while holding the other fixed, which no observational study of a production log can do and a controlled replay of a released benchmark matrix can.

**Contributions.**

1. A decomposition (Prop. 3.1) that separates the two failures and, more usefully, states the condition under which the standard repair is valid. The randomized-audit repair works only if the evaluator's average error on the router's requests matches its average error on the general population. Using the same judge on both is *not* sufficient.
2. A crossed experiment that switches each failure on and off independently on the same data, holding model roster, requests, contexts and log volume fixed, replicated end to end on a second released matrix with a different roster.
3. Findings a practitioner can act on, covering what an audit repairs and what it cannot, why deconfounding the surface the budget is spent on matters more than the acquisition rule but not without limit, a usable test for whether an audit is worth buying, and the loss asymmetry at which a deployment gate starts to pay.

## 2 Setup and notation

Consider a team that sends routine requests to a cheaper model and difficult requests to a stronger one. We label each request by $i$ and group similar requests into $J$ contexts, where each context represents a request type such as routine questions or difficult coding questions. The contexts are numbered $j = 1, \ldots, J$, and $Z_i \in \{1, \ldots, J\}$ records the context of request $i$. The team compares $K$ candidate models, numbered $k = 1, \ldots, K$. For each logged request $i$, $A_i \in \{1, \ldots, K\}$ records which model answered it. Pairing model $k$ with context $j$ gives a model and context cell, written as $(k, j)$, so there are $K \times J$ cells in total. A deployment policy $\pi$ chooses one model for each context. Thus $\pi(j) = k$ means that the policy sends requests in context $j$ to model $k$, so three models across four request types give twelve cells.

Table 1: Notation. The three surfaces differ only in *who was measured* and *what the measurement was*, and keeping them apart is the whole argument.

| Symbol | Name | What it is |
|---|---|---|
| $\mu^{\mathrm{ref}}_{kj}$ | target surface | average designated outcome of model $k$ on context-$j$ requests |
| $M_{kj}$ | logged surface | average *recorded score* of model $k$ over the context-$j$ requests *the router sent it* |
| $R_{kj}$ | audit readout | average recorded score of model $k$ on a fresh *random* sample of context-$j$ requests |
| $\mu^{\mathrm{op}}_{kj}$ | population-operational surface | what the evaluator would report for model $k$ on *all* context-$j$ requests, with selection removed and evaluator error kept |
| $B^{\mathrm{sel}}_{kj}$ | selection bias | how much easier or harder the router's requests were, in target units |
| $C_{kj}$ | evaluator distortion | how far the recorded score sits from the target outcome, per cell |
| $\Delta_j$ | target margin | gap between best and second-best model in context $j$, under $\mu^{\mathrm{ref}}$ |
| $R^{\mathrm{ref}}(\pi)$ | deployment regret | target value lost by deploying $\pi$ instead of $\pi^{\mathrm{ref}}$, averaged over contexts |

**The outcome the decision is about.** Suppose the team wants to choose the model that provides the greatest value for each type of request. We call this value the target outcome. For request $i$ and model $k$, $Y^{\mathrm{ref}}_i(k)$ is the target outcome that model $k$ would receive if it answered request $i$. In our experiments this is a released benchmark score, and in a deployment it could be any measure fixed in advance, such as correctness or user satisfaction. For model $k$ and context $j$, $\mu^{\mathrm{ref}}_{kj} = \mathbb{E}[Y^{\mathrm{ref}}_i(k) \mid Z_i = j]$ is the model's average target outcome on requests in that context. The target policy, written $\pi^{\mathrm{ref}}$, assigns each context to the model with the highest average target outcome. Thus $\pi^{\mathrm{ref}}(j) = \arg\max_k \mu^{\mathrm{ref}}_{kj}$. We call this a designated target rather than ground truth because it is the objective the team has chosen, and all results below ask whether the logged evaluation recovers the decisions that objective implies.

**Three streams, one evaluator.** The same evaluator scores requests in three data sources. The observational log ($s = o$) contains requests after the router has chosen which model answers them, the randomized audit ($s = i$) scores models on randomly selected requests, and the held-out verification set ($s = v$) is kept apart and used only to check a final deployment decision. For request $i$, model $k$, and data source $s$, $\widetilde{Y}^s_i(k)$ is the score reported by the evaluator. Its error is the difference between this score and the target outcome, $\widetilde{Y}^s_i(k) - Y^{\mathrm{ref}}_i(k)$. Even when the same evaluator is used, its average error may differ between the log and the audit because the router chooses which requests each model receives in the log. For model $k$ and context $j$, $C^{o,\mathrm{sel}}_{kj}$ is the evaluator's average error on the requests in context $j$ that the router sent to model $k$, while $C^{i,\mathrm{pop}}_{kj}$ is its average error when model $k$ is evaluated on a random sample of all requests in that context. Formally,

$$C^{o,\mathrm{sel}}_{kj} = \mathbb{E}\Big[\widetilde{Y}^o_i(k) - Y^{\mathrm{ref}}_i(k) \mid A_i{=}k, Z_i{=}j\Big], \qquad C^{i,\mathrm{pop}}_{kj} = \mathbb{E}\Big[\widetilde{Y}^i_i(k) - Y^{\mathrm{ref}}_i(k) \mid Z_i{=}j\Big].$$

Finally, $B^{\mathrm{sel}}_{kj} = \mathbb{E}[Y^{\mathrm{ref}}_i(k) \mid A_i{=}k, Z_i{=}j] - \mu^{\mathrm{ref}}_{kj}$ measures the selection bias created by the router, comparing how well model $k$ would perform on the requests it was assigned with how well it would perform on all requests in context $j$. A positive value means the router gave it more favorable requests and a negative value harder ones. Table 1 summarizes the notation.

## 3 What a randomized audit can and cannot repair

A randomized audit gives each model a fairer sample of requests, so it can help separate model quality from the router's assignment choices, but the correction is valid only under a specific condition. Proposition 3.1 splits the difference between the logged score and the audit score into two parts, the selection bias created by the router and any change in the evaluator's average error between the routed and the randomly sampled requests. The audit recovers selection bias only when the second part is zero.

**Proposition 3.1** (Validity decomposition)**.** *Up to sampling noise,* $M_{kj} - R_{kj} = B^{\mathrm{sel}}_{kj} + \big(C^{o,\mathrm{sel}}_{kj} - C^{i,\mathrm{pop}}_{kj}\big)$. *Hence* (a) *the logged-minus-audit difference recovers* $B^{\mathrm{sel}}_{kj}$ *exactly when* $C^{o,\mathrm{sel}}_{kj} = C^{i,\mathrm{pop}}_{kj}$*, a*

*condition we call* assignment invariance*, meaning that the evaluator's mean error on the requests the router sent to model $k$ equals its mean error on all context-$j$ requests.* (b) *A sufficient condition is that the same evaluator scores both streams* and *its error depends on a request only through its cell, so that $C_{kj}^{o,\mathrm{sel}} = C_{kj}^{i,\mathrm{pop}} = C_{kj}$. Subtracting the recovered difference from $M$ then gives a surface targeting $\mu_{kj}^{\mathrm{op}} = \mu_{kj}^{\mathrm{ref}} + C_{kj}$ rather than $\mu_{kj}^{\mathrm{ref}}$. In particular a context-invariant model preference $C_{kj} = C_k$ cancels from the difference entirely and therefore* survives *in the corrected surface.*

*Remark* 3.2 (Why using the same evaluator is not enough). If the evaluator makes different kinds of errors on easy and difficult requests, and the router sends these requests to models at different rates, then its average error can still differ across the two samples. Even when it does hold, the audit removes only assignment bias, and any systematic error in what the evaluator rewards remains in the corrected comparison. Our simulated distortion is cell-constant by construction, so part (b) applies. Clipping to a bounded score range makes the *realised* error slightly request-dependent, so we measure that residual rather than assume it away (§5, App. B).

**Corollary 3.3** (When the corrected surface still names the best model)**.** *Let the deployed surface in context $j$ be $\mu_{kj}^{\mathrm{ref}} + \varepsilon_{kj}$, where $\varepsilon_{kj}$ collects leftover evaluator distortion, correction error and readout noise, and let $\Delta_j = \mu_{\pi^{\mathrm{ref}}(j),j}^{\mathrm{ref}} - \max_{k \neq \pi^{\mathrm{ref}}(j)} \mu_{kj}^{\mathrm{ref}}$. The deployed policy still picks $\pi^{\mathrm{ref}}(j)$ whenever $\max_k(\varepsilon_{kj} - \varepsilon_{\pi^{\mathrm{ref}}(j),j}) < \Delta_j$. A sufficient condition is $\max_k |\varepsilon_{kj}| < \Delta_j/2$.*

These errors change the selected model only where the best two are close in target value. Proofs are in App. A.

**The four experimental conditions.** We study the two failures separately and together. The first condition has neither failure. Models receive comparable requests and the evaluator measures the declared target. The second has assignment failure only. The router sends different kinds of requests to different models, but the evaluator remains faithful. The third has measurement failure only. Models receive comparable requests, but the evaluator systematically distorts their scores. The fourth contains both failures. We refer to these conditions as neither, assignment only, measurement only, and both. In figures and tables, A0 and A1 mark the absence or presence of assignment failure and D0 and D1 the absence or presence of evaluator distortion. We also include A1–D2, where the evaluator behaves differently in the log and the audit, as a diagnostic. Every condition can still contain ranking errors from limited sample sizes, including the one with neither validity failure.

# 4 The audit protocol

Buying a cell means running that model on $n{=}30$ randomly drawn context-$j$ requests and averaging, which gives an unbiased finite-sample estimate of $R_{kj}$. Given a budget of $B$ cells:

1. **Correct.** From the cells bought so far fit one offset per model, $b_k = \mathrm{mean}_{j\ \mathrm{bought}}(M_{kj} - R_{kj})$, and deploy $V_{kj} = M_{kj} - b_k$, overwriting bought cells with their readouts. We call $V$ the *deconfounded surface*.
2. **Allocate.** Buy next the unbought cell in the context whose *deconfounded* top-two margin is smallest, which is the decision the current surface leaves unresolved.
3. **Re-correct.** Refit $b$ after every readout and repeat.
4. **Check (optional).** Reserve $m_v$ of the $B$ cells for a *held-out decision check* on requests no other stage touched. Buy the cells the status-quo and candidate policies disagree about, in matched pairs, and switch only if the candidate wins.

Steps 1 to 3 form a deliberately simple reference protocol, which we call CASE. It sits in the family that combines a confounded log with a small unbiased sample [Athey et al., 2020, Kallus et al., 2018, Rosenman et al., 2018, Agarwal et al., 2023] and in budgeted causal experimental design [Gao et al., 2026b, 2025]. We make no claim for the margin heuristic, so we also run a decision-information rule (R-EPIG, Gao et al., 2026a) on both surfaces, and §6 shows where each is adequate. Step 4 is a conditional gate rather than a safe-deployment certificate, because it bounds harm relative to the status quo only on the cells it checks, under the evaluator that checks them.

# 5 Experimental design

**Two corpora.** The primary study replays the LLMRouterBench performance–cost track [Li et al., 2026], which holds $11{,}289$ requests aligned across five benchmark slices and five models spanning the cost–capability range. Every conclusion is re-derived by running the identical pipeline on RouterBench [Hu et al., 2024], which holds $36{,}497$ requests, $11$ models and five benchmark families, and shares no requests, models or scoring convention with the first. Both are offline replays rather than production traces, so neither carries temporal drift, dynamic router updates, retries or missing counterfactual outputs (§7). Contexts are $J{=}8$ frozen KMeans clusters of request embeddings.

**Cohorts.** Each request has exactly one role, either router-train ($20\%$), observational log ($50\%$), audit ($12\%$), verification ($8\%$) or scoring ($10\%$), stratified by slice. Contexts, the router and every evaluator-design constant are fit on router-train only. Nothing except final scoring and the two labelled oracles reads the scoring cohort, and a leakage test (App. B) confirms that replacing its outcomes with random values leaves the acquired cells, the fitted correction, the policy and the gate decision bit-identical.

**Factor A (assignment).** **A1** routes with a FrugalGPT-style cost cascade [Chen et al., 2023] fit on router-train, with exploration rate $0.15$. **A0** takes the *same* logged requests and re-partitions each context's requests uniformly at random into blocks of *exactly* the same per-cell sizes, so selection bias is zero while log volume, per-cell counts and evaluator are unchanged.

**Factor D (evaluator distortion).** **D0** has no distortion. **D1** applies a stable model-by-context distortion shared by all three streams, $C_{kj} = \lambda z_k s_j$, where $z_k$ is the standardized model capability rank and $s_j \in \{\pm 1\}$ splits harder from easier contexts, both computed on router-train only, and $\lambda = 0.5 \times \mathrm{median}_j \Delta_{j,\mathrm{train}}$ is fixed by formula before any run. Scores are clipped to $[0, 1]$, which attenuates the distortion to an effective shift of $0.0100_{\pm 0.0007}$ against a median target margin of $0.041$, and also makes the realised error slightly request-dependent, so assignment invariance holds only approximately. The measured residual is $0.0018_{\pm 0.0001}$, twice which is an order of magnitude inside the $0.041$ margin Cor. 3.3 requires, so it cannot flip a typical context. **D2** keeps A1 and halves the distortion on the audit stream.

**Methods, outcomes, replication.** The arms are trusting the log, a uniform audit, CASE, R-EPIG on the observed and on the deconfounded surface, and two labelled oracles. The *population-operational oracle* $\mu^{\mathrm{op}}$ removes selection exactly while keeping evaluator error, so it is the ceiling for a perfect assignment repair, and the *reference-outcome oracle* is $\mu^{\mathrm{ref}}$. The confirmatory primary outcome is deployment regret $R^{\mathrm{ref}}$ on the untouched scoring cohort, and the second is the false-switch rate. Budgets are $15$, $25$ and $35\%$ of supported cells, which is $B{=}13$ cells or $390$ audited requests at $35\%$ on the primary corpus. We run $30$ prompt-cohort partitions crossed with $6$ acquisition seeds, average seeds *within* a partition first, and report $\pm$ as a paired standard error over partitions. A partition resamples cohorts, contexts and the router within one fixed corpus, so it is not an independent draw of new requests. App. D therefore gives $95\%$ intervals under partition, request and slice resampling, and we quote the last wherever it changes a conclusion.

# 6 Results

We first test the paper's central claims. We ask whether a router-generated log identifies the best model, how much a randomized audit repairs the resulting assignment bias, what determines how effectively the budget is used, and whether assignment correction also repairs measurement error. We then ask whether a held-out check can prevent harmful deployment changes and whether the first few randomized measurements reveal when a larger audit is worth conducting. Throughout we report deployment regret, the value lost by selecting the wrong model, and identification error, how often the selected model differs from the target-best one.

## 6.1 Core validity results

**Q1. The log names the wrong model in two thirds of request types.** Under A1–D0 the logged surface picks a model other than the target-best one in $67.5\%_{\pm 2.4}$ of contexts, against $39.6\%_{\pm 2.6}$

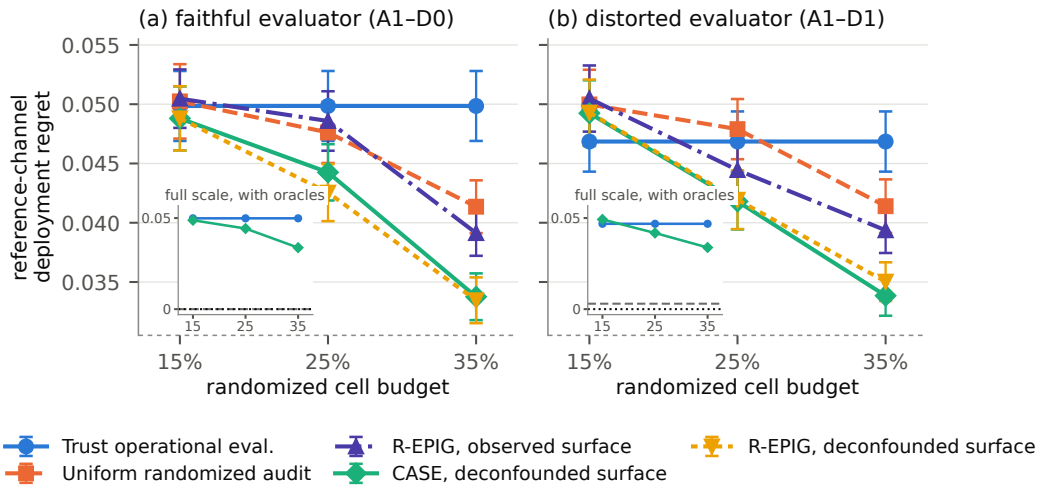



Figure 1: **Deconfounding the acquisition surface matters more than the rule that spends the budget.** Deployment regret (lower is better) under the selective-router log, with error bars giving paired standard errors over 30 partitions. The two deconfounded-surface arms track each other and sit below the surface-agnostic uniform rule and the observed-surface rule, and they buy 86% of the same cells (Q3). **The main axis is truncated** so that the methods, which span 0.033 to 0.051, are legible, and the inset in each panel repeats trust and CASE on the full axis with both oracles, so the spread between the methods can be read against the much larger gap that separates all of them from a perfect repair. Per-slice values appear in Fig. 3 and Table 4.

for the matched-volume common-population log. In regret the figures are $0.0499_{\pm 0.0030}$ against $0.0214_{\pm 0.0024}$. The gap is selection rather than sample size, because the two logs hold the same requests in the same per-cell counts, and no outcome oracle enters A1. On RouterBench the same statistics are $46.9\%_{\pm 3.3}$ and $25.6\%_{\pm 2.9}$.

**Q2. Auditing repairs a substantial fraction, and no more.** At a 35% budget under A1–D0, CASE lowers regret from $0.0499_{\pm 0.0030}$ to $0.0338_{\pm 0.0020}$, a paired difference of $-0.0161$ with 95% CI $[-0.0219, -0.0103]$, better on 27 of 30 partitions and excluding zero under all three resampling schemes of App. D. It improves on all five slices, and on RouterBench the effect is larger, from 0.0784 to 0.0421, a paired difference of $-0.0363_{\pm 0.0042}$ better on *every* partition. How far this goes towards a perfect repair depends on the estimand, so we give both. The mean over partitions of the per-partition share of the trust-to-oracle gap that CASE closes is 28%, and the share formed from aggregate means is 32%.

*Why only a third?* Not the budget. Prop. 3.1 is cellwise but the deployed correction is one offset per model, and the router makes that misspecified, since $68\%_{\pm 2}$ of the selection distortion is not constant across contexts. Replacing the budget-limited fit with the *best-fitting member of the same family*, computed from the truth with no budget at all, reaches $0.0318_{\pm 0.0025}$, only $0.0020_{\pm 0.0024}$ below CASE with 95% CI $[-0.0030, +0.0070]$. So of the trust-to-oracle gap, CASE closes the 28% above, a further 5% is reachable inside the action-level family (14% at the top of that interval), and roughly two thirds lies outside it, all three shares being mean-of-ratios over partitions. More readouts would not help, whereas a richer corrector would, but the richer families we tried overfit 13 purchased cells (App. C).

**Q3. The surface decides more than the rule, but the rule is not free.** Crossing the acquisition rule (smallest-margin against decision-information) with the acquisition surface (observed against deconfounded) separates the two. *Holding the rule fixed, moving to the deconfounded surface helps in all eight cells across both corpora*, by 0.0036 to 0.0115. Holding the surface fixed, the rule ties on the five-model corpus, with a paired difference of $+0.0003_{\pm 0.0011}$ and two one-sided tests rejecting a difference larger than 0.005 at $p = 10^{-4}$. The two rules also buy $86.5\%_{\pm 1.0}$ of the same cells, against $61.4\%_{\pm 1.0}$ when the same rule moves to the observed surface, so the tie is mechanical rather than coincidental. **On the eleven-model corpus the rule does matter, and the margin heuristic loses to**

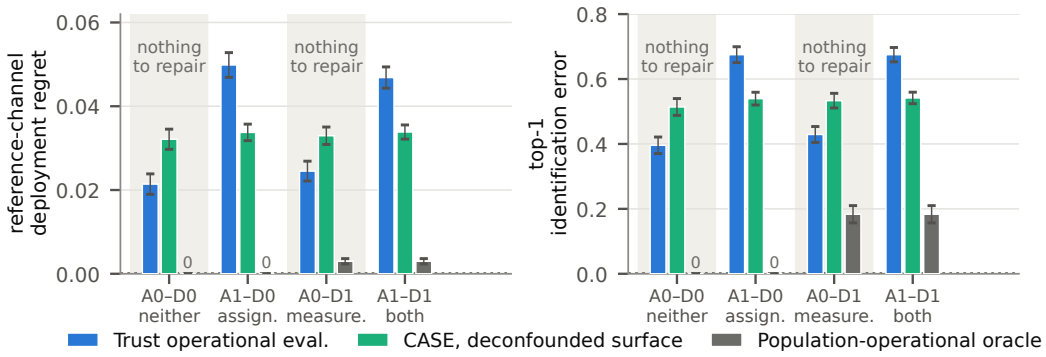



Figure 2: **The four-cell decomposition** at a 35% budget, where the reference-outcome oracle is 0 in every cell by construction. Shaded cells have no assignment invalidity to repair, so CASE sitting *above* trust there is the variance cost of buying 30-request readouts rather than a failure of the method (Q6). In the both-failures cell, the part of the logged loss a cellwise-perfect repair removes is $G_{\text{sel}} = 0.0439_{\pm 0.0029}$, and the measurement floor that survives it is $G_{\text{meas}} = 0.0030$ in regret but $18.3\%_{\pm 2.7}$ in identification error.

**the decision-information rule by $+0.0140_{\pm 0.0021}$.** The bounded claim is that deconfounding the surface is the robust lever and that a simple margin rule becomes inadequate as the roster grows.

**Q4. None of it buys measurement validity.** In the both-failures cell the logged evaluation loses $0.0469_{\pm 0.0025}$. A cellwise-perfect repair removes $G_{\text{sel}} = 0.0439_{\pm 0.0029}$ and leaves $G_{\text{meas}} = 0.0030$, with 95% CI $[+0.0017, +0.0043]$ over partitions, though it is *not* separable from zero when the five slices are resampled as clusters (App. D). The measurement-only cell makes the point with no selection bias to repair, since CASE sits at $0.0330_{\pm 0.0021}$ against $0.0030_{\pm 0.0006}$ for the population-operational oracle.

The floor is small in *value* and large in *identification*. With selection bias removed exactly, the evaluator still names the wrong model in $18.3\%_{\pm 2.7}$ of contexts on the primary corpus and $13.8\%_{\pm 1.5}$ on RouterBench, both at the pre-specified distortion strength and $J{=}8$. Across every sensitivity we ran the floor spans 10.0 to 30.6%, at its lowest under a context-invariant model preference and its highest at twice the pre-specified distortion (App. C), so the headline pair is a point inside a range rather than a constant. Reversals concentrate where the target margin is smallest, so they are cheap in regret and expensive in ranking, and the two reports disagree by an order of magnitude about what this evaluator's bias costs. **We stress what this number is and is not.** Applying Cor. 3.3 to the margin distribution alone, with no reference to which model actually wins, predicts $28.8\%_{\pm 2.7}$ against the $18.3\%_{\pm 2.7}$ observed, so the rate is largely fixed by $\lambda$ and the margins and carries *no* information about how biased any deployed judge is. What is not arithmetic is the structural claim it illustrates, that a cellwise-perfect assignment repair leaves this error untouched.

### 6.2 Practical deployment questions

**Q5. A same-evaluator check cuts harm but cannot validate the construct, and gating pays only under an asymmetry.** Both held-out checks cut harmful switches, from $0.178_{\pm 0.030}$ ungated to $0.061_{\pm 0.019}$ for the same evaluator and $0.072_{\pm 0.021}$ for the reference channel (Table 5). **We had pre-specified that the same-evaluator check would leave *more* harmful switches, and it does not.** It is correlated with the baseline it checks, because the log-derived policy is the argmax of the operational surface, so reading verification through the same distortion tilts toward retention. At the stronger bias level it switches less (0.55 against 0.62) and falsely retains more ($0.242_{\pm 0.035}$ against $0.208_{\pm 0.032}$). What it provably cannot do is get the verdict right. On 4.6% of contested contexts, or 8.3% at the stronger level, its *population* comparison points the opposite way from the target's, a property of the channel rather than the sample, unchanged by budget and flat in readout size.

Gating is not free. It trades 0.117 fewer false switches for 0.233 more false retentions, so it lowers expected loss only if a harmful switch costs at least $2.0\times$ a missed improvement at the 35% budget,

Table 2: Validity scope of a randomized audit over a router-generated log.

| Threat | Addressed? | Required condition |
|---|---|---|
| Router-induced selective assignment | Yes, partly | Randomizable cells, correction family rich enough |
| Finite audit noise | Partially | Readouts no thinner than the log cells they replace (Q6) |
| Context-varying selection residual | **Largely not** | Two thirds of it lies outside the fitted family |
| Stable shared evaluator distortion | **No** | Construct validation or a *validated* channel |
| Evaluator error that depends on the request | Can contaminate | Assignment invariance (Prop. 3.1a) |
| Unsupported model–context cells | No | Exploration or expanded support |

and $1.0\times$ at the $15\%$ budget, where the audit behind it is too thin to trust (App. C). *State the asymmetry before quoting the gate.*

**Q6. You can tell in advance whether the audit is worth buying.** Where the log is already common-population and its cells are larger than the readouts that would replace them, auditing *raises* regret, from $0.0214_{\pm 0.0024}$ to $0.0321_{\pm 0.0024}$ (paired $+0.0107$, CI $[+0.0067, +0.0147]$, not separable from zero under slice resampling), because each 30-request readout is noisier than the $\approx$140-request log cell it overwrites. The binding condition is that density comparison rather than the audit itself, and on RouterBench, whose log cells are far larger, the cost disappears ($-0.0005_{\pm 0.0037}$). A test makes this actionable. After just the $K$ per-model seed readouts, take the dispersion of the logged-minus-audit differences, standardised by the noise they would carry under a common-population log. It averages $1.06_{\pm 0.03}$ under A0 and $1.28_{\pm 0.04}$ under A1, with AUC $0.79$, and auditing only above a threshold gives mean regret $0.0313$ against $0.0329$ for always auditing and $0.0356$ for never auditing. The threshold is fitted in-sample, so the AUC is the honest separation, but five cells can inform the decision rather than leaving it blind.

# 7 Conclusion

A logged model comparison can fail in two ways that call for different remedies. Randomized auditing repairs the assignment failure under stated conditions, removing between a quarter and a third of the distance to a perfect repair on the primary corpus and rather more on the second. It does not repair the measurement failure at all. Once selection bias is removed exactly, the evaluator still names the wrong model in $14$ to $18\%$ of request types on the two corpora at the pre-specified distortion strength, and in $10$ to $31\%$ across our sensitivity analyses, in every case whatever the audit budget. Three further results are practical. What makes a budget productive is the surface it is spent on rather than the rule that spends it, though the rule regains importance as the roster grows. A held-out check reduces harmful deployment changes but inherits the construct it was meant to check, and pays only when a harmful switch costs roughly twice a missed improvement. And the first few randomized measurements already indicate whether a larger audit is worth buying, which matters where the readouts would be thinner than the log cells they overwrite. That cost, $+0.0107$ on the primary corpus and not separable from zero once benchmark slices are resampled as clusters, is a statement about readout size relative to log-cell density rather than about auditing as such, and it disappears on RouterBench, whose log cells are far larger.

**Limitations.** Table 2 lists what the protocol addresses, and two limitations matter most. The identification result of §3 holds for any evaluator, but the magnitudes come from a pre-specified synthetic distortion rather than a measured judge, so nothing here estimates construct bias for a deployed LLM judge, and measuring that is the natural next step. The reference channel reads the designated target during verification, an experimental control rather than something a production team can obtain on demand, so it bounds what a better channel could achieve rather than describing an available one.

**A reporting checklist.** A comparison read off a router-generated log should report (i) the assignment mechanism and the per-cell logged counts, (ii) which evaluator scored each stream and whether its error is plausibly invariant to routing, not merely that the same judge was used, (iii) the outcome the conclusion is *about* and how it differs from what was measured, (iv) the target margin per context beside the ranking, so Cor. 3.3 is checkable, and (v) both a value and an identification metric, which can disagree by an order of magnitude.

# References


Anish Agarwal, Munther Dahleh, Devavrat Shah, and Dennis Shen. Causal matrix completion. In *Conference on Learning Theory (COLT)*, 2023.

Susan Athey, Raj Chetty, Guido Imbens, and Hyunseung Kang. Combining experimental and observational data to estimate treatment effects on long term outcomes. *arXiv preprint arXiv:2006.09676*, 2020.

Taehun Cha and Donghun Lee. ABC3: Active bayesian causal inference with cohn criteria in randomized experiments. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N. Angelopoulos, Tianle Li, Dacheng Li, Banghua Zhu, Hao Zhang, Michael I. Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot arena: An open platform for evaluating LLMs by human preference. In *International Conference on Machine Learning (ICML)*, 2024.

Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-aware query routing. In *International Conference on Learning Representations (ICLR)*, 2024.

Tao Feng, Yanzhen Shen, and Jiaxuan You. GraphRouter: A graph-based router for LLM selections. In *International Conference on Learning Representations (ICLR)*, 2025.

Erdun Gao, Jake Fawkes, and Dino Sejdinovic. Causal-EPIG: A prediction-oriented active learning framework for CATE estimation. *arXiv preprint arXiv:2509.21866*, 2025.

Erdun Gao, Liang Zhang, Jake Fawkes, Aoqi Zuo, Wenqin Liu, Haoxuan Li, Mingming Gong, and Dino Sejdinovic. Observationally informed adaptive causal experimental design. *arXiv preprint arXiv:2603.03785*, 2026a. To appear, KDD 2026.

Jiacan Gao, Xinyan Su, Mingyuan Ma, Yiyan Huang, Xiao Xu, Xinrui Wan, Tianqi Gu, Enyun Yu, Jiecheng Guo, and Zhiheng Zhang. Budgeted active experimentation for treatment effect estimation from observational and randomized data. *arXiv preprint arXiv:2602.22021*, 2026b.

Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt Keutzer, and Shriyash Kaustubh Upadhyay. RouterBench: A benchmark for multi-LLM routing system. *arXiv preprint arXiv:2403.12031*, 2024.

Abigail Z. Jacobs and Hanna Wallach. Measurement and fairness. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 375–385, 2021.

Nathan Kallus, Aahlad Manas Puli, and Uri Shalit. Removing hidden confounding by experimental grounding. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Guannan Lai, Haoran Hu, Long Chen, Zhenguo Li, and Han-Jia Ye. From sampled outcomes to capability distributions: Rethinking supervision for LLM routing. *arXiv preprint arXiv:2606.06924*, 2026.

Hao Li, Yiqun Zhang, Zhaoyan Guo, Chenxu Wang, Shengji Tang, Qiaosheng Zhang, Yang Chen, Biqing Qi, Peng Ye, Lei Bai, Zhen Wang, and Shuyue Hu. LLMRouterBench: A massive benchmark and unified framework for LLM routing. *arXiv preprint arXiv:2601.07206*, 2026.

Mingxuan Li, Junzhe Zhang, and Elias Bareinboim. Confounding robust deep reinforcement learning: A causal approach. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL `https://arxiv.org/abs/2510.21110`. Causal DQN: offline confounded data + online interventions.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data. In *International Conference on Learning Representations (ICLR)*, 2025.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Inioluwa Deborah Raji, Emily M. Bender, Amandalynne Paullada, Emily Denton, and Alex Hanna. AI and the everything in the whole wide world benchmark. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

Evan Rosenman, Art B. Owen, Michael Baiocchi, and Hailey Banack. Combining observational and experimental data to find heterogeneous treatment effects. *arXiv preprint arXiv:1804.05857*, 2018.

Jinwu Su, Feng Lin, Zhengyu Feng, Hang Zheng, Tianyi Wang, Zheng Xiao, and Hui Wang. CP-Router: An uncertainty-aware router between LLM and LRM. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2026.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 9440–9450, 2024.

Xuelin Yang, Licong Lin, Susan Athey, Michael I. Jordan, and Guido W. Imbens. Cross-validated causal inference: A modern method to combine experimental and observational data. *arXiv preprint arXiv:2511.00727*, 2025.

Haozhen Zhang, Tao Feng, and Jiaxuan You. Router-R1: Teaching LLMs multi-round routing and aggregation via reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

Hang Zheng, Hongshen Xu, Yongkai Lin, Shuai Fan, Lu Chen, and Kai Yu. DiSRouter: Distributed self-routing for LLM selections. In *International Conference on Learning Representations (ICLR)*, 2026. URL `https://openreview.net/pdf?id=KDcwXKr0NU`.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023.

# A Proofs

*Proof of Prop. 3.1.* Write the operational score as the target plus an error, $\widetilde{Y}_i^s(k) = Y_i^{\mathrm{ref}}(k) + e_i^s(k)$. Conditioning on the logged assignment and adding and subtracting $\mu_{kj}^{\mathrm{ref}}$,

$$M_{kj} = \mathbb{E}\big[Y_i^{\mathrm{ref}}(k) \mid A_i{=}k, Z_i{=}j\big] + \mathbb{E}[e_i^o(k) \mid A_i{=}k, Z_i{=}j] = \mu_{kj}^{\mathrm{ref}} + B_{kj}^{\mathrm{sel}} + C_{kj}^{o,\mathrm{sel}}.$$

The audit readout draws context-$j$ requests at random, so $R_{kj} = \mu_{kj}^{\mathrm{ref}} + C_{kj}^{i,\mathrm{pop}}$. Subtracting gives $M_{kj} - R_{kj} = B_{kj}^{\mathrm{sel}} + (C_{kj}^{o,\mathrm{sel}} - C_{kj}^{i,\mathrm{pop}})$, which recovers $B_{kj}^{\mathrm{sel}}$ if and only if the bracket vanishes, giving part (a). Note what (a) does *not* say. Using the same evaluator on both streams equates the two error *functions*, $e^o = e^i$, but not the two conditional means, and those means are taken over different populations whenever the router selects within a context.

For (b), suppose additionally that the error depends on the request only through its cell, so that $\mathbb{E}[e_i^s(k) \mid \mathcal{E}, Z_i{=}j] = C_{kj}$ for every conditioning event $\mathcal{E}$ within context $j$. Then $C_{kj}^{o,\mathrm{sel}} = C_{kj}^{i,\mathrm{pop}} = C_{kj}$, the recovered difference is $B_{kj}^{\mathrm{sel}}$, and removing it from $M_{kj} = \mu_{kj}^{\mathrm{ref}} + B_{kj}^{\mathrm{sel}} + C_{kj}$ leaves $\mu_{kj}^{\mathrm{ref}} + C_{kj} = \mu_{kj}^{\mathrm{op}}$. If moreover $C_{kj} = C_k$ for all $j$, then $C_k$ never enters $M_{kj} - R_{kj}$ at all, so no differencing estimator can see it while $\mu_{kj}^{\mathrm{op}}$ retains it, and a per-model correction fitted from the differences therefore cannot remove it. App. C instantiates exactly this case. □

*Proof of Cor. 3.3.* Write $k^\star = \pi^{\mathrm{ref}}(j)$. The deployed policy picks $k^\star$ if and only if for every $k \neq k^\star$, $\mu_{k^\star j}^{\mathrm{ref}} + \varepsilon_{k^\star j} > \mu_{kj}^{\mathrm{ref}} + \varepsilon_{kj}$, that is, $\varepsilon_{kj} - \varepsilon_{k^\star j} < \mu_{k^\star j}^{\mathrm{ref}} - \mu_{kj}^{\mathrm{ref}}$. The right-hand side is at least $\Delta_j$, which gives the stated condition, and $\varepsilon_{kj} - \varepsilon_{k^\star j} \le 2\max_k |\varepsilon_{kj}|$ gives the sufficient form. □

*Remark* A.1 (Why the reversals are cheap). Cor. 3.3 implies that with distortion bounded by $\epsilon$, only contexts with $\Delta_j < 2\epsilon$ can flip, and each flip costs at most $\Delta_j < 2\epsilon$. This is the pattern §6 reports, in which many contexts flip but the regret cost is an order of magnitude below a distortion-sized loss per flipped context.

# B Leakage and algebra tests

**Reproducibility.** Anonymized supplementary material accompanies the submission, containing the experiment package, the pre-registered analysis plan, all tests below, and the long-form result files behind every number. It regenerates in minutes and uses only released per-request benchmark outputs, with no new model inference. The result files key the evaluator factor E where the paper writes D, and the mapping is recorded in the supplementary claim registry.

All six tests run before any result file is written, and the runner exits non-zero if one fails. Values are from partition 0 of the primary corpus, and all 30 partitions pass.

1. **Cohort isolation.** The five cohorts are pairwise disjoint and exhaust the 11,289 requests.
2. **Assignment matching.** A paired A0/A1 run has identical per-cell counts, identical total log volume (5,644 requests) and identical evaluator distortion, and every A0 cell draws only from its own context's log cohort.
3. **Reference-channel leakage.** Replacing the scoring cohort's target outcomes with uniform random values leaves the acquired cells, the fitted correction, the deployed surface (max difference 0.0), the candidate policy and the gate decision identical. The allocator is called with the scoring surface set to `None`, so reading it would raise rather than bias.
4. **Stable-evaluator algebra.** With $C^o = C^i$ the logged-minus-audit difference is unchanged by the distortion (mean absolute change 0.0019 against an effective distortion of 0.0121), while the population-operational surface differs from the target surface by exactly that distortion. Clipping attenuates the nominal distortion by $0.476{\pm}0.003$.
5. **Assignment invariance** (Prop. 3.1(a)). The residual $|C^{o,\mathrm{sel}} - C^{i,\mathrm{pop}}|$ is measured rather than assumed. It is $0.0018{\pm}0.0001$ over 30 partitions, against a cohort-sampling floor of $0.0013{\pm}0.0001$ with routing removed and a selection bias of $0.0991{\pm}0.0019$, so contamination is $1.8\%{\pm}0.1$. Twice the residual is 0.0036, an order of magnitude below the 0.041 median target margin, so by Cor. 3.3 it cannot flip a typical context. It is non-zero because clipping makes the realised per-request error depend on the request's own target outcome, which the router selects on.

6. **Stream-mismatch algebra.** With $C^i = \frac{1}{2}C^o$ the estimated cellwise difference gains the predicted extra term, with correlation $0.94$ against $\frac{1}{2}C^o$ and slope $0.46$, the clipping attenuation.

# C Additional results

Each subsection sets out the question that motivated the analysis, the factor that was varied, and the finding, followed by the supporting table or figure.

## C.1 The four validity cells in one table

The main text quotes individual cells of the factorial, but a reader checking the decomposition needs all four together, with every arm, on one scale. Table 3 gives all six arms at the $35\%$ budget in each of the four cells, over 30 partitions crossed with 6 acquisition seeds. The pattern the paper argues for is then visible in one place. Auditing helps only where assignment invalidity is present, which is the contrast between the A1 rows and the A0 rows, and the population-operational oracle is zero under a faithful evaluator but not under a distorted one.

Table 3: Deployment regret at a $35\%$ budget, all four validity cells, primary corpus, with $\pm$ a paired SE over 30 partitions.

| Cell | Trust | Uniform | CASE | R-EPIG$_{\text{corr}}$ | Pop.-op. oracle | Ref. oracle |
|---|---|---|---|---|---|---|
| A0–D0 | $0.0214_{\pm 0.0024}$ | $0.0338_{\pm 0.0021}$ | $0.0321_{\pm 0.0024}$ | $0.0326_{\pm 0.0024}$ | $0.0000_{\pm 0.0000}$ | $0.0000_{\pm 0.0000}$ |
| A1–D0 | $0.0499_{\pm 0.0030}$ | $0.0414_{\pm 0.0022}$ | $0.0338_{\pm 0.0020}$ | $0.0335_{\pm 0.0019}$ | $0.0000_{\pm 0.0000}$ | $0.0000_{\pm 0.0000}$ |
| A0–D1 | $0.0245_{\pm 0.0024}$ | $0.0326_{\pm 0.0019}$ | $0.0330_{\pm 0.0021}$ | $0.0325_{\pm 0.0019}$ | $0.0030_{\pm 0.0006}$ | $0.0000_{\pm 0.0000}$ |
| A1–D1 | $0.0469_{\pm 0.0025}$ | $0.0414_{\pm 0.0023}$ | $0.0338_{\pm 0.0017}$ | $0.0350_{\pm 0.0016}$ | $0.0030_{\pm 0.0006}$ | $0.0000_{\pm 0.0000}$ |

## C.2 Does the result come from one benchmark slice?

A pooled improvement can be produced by a single slice while the others are flat or worse, in which case it would not be a general finding. To check this we re-score the pooled deployed policy on each slice's scoring requests separately, against the pooled target policy. The audit improves on all five slices rather than on one, although the size varies by roughly a factor of ten between MMLU-Pro and ArenaHard, which is why the slice-cluster interval in App. D is much wider than the partition interval.

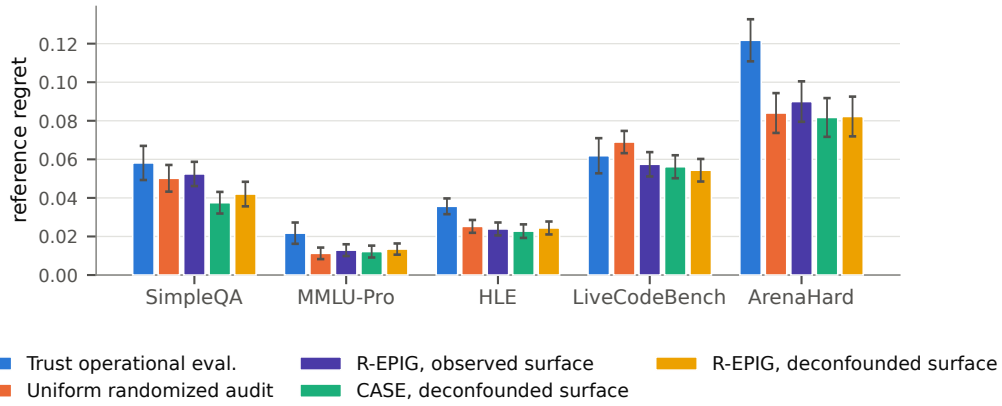



Figure 3: Slice contributions to the pooled-policy regret at a $35\%$ budget under A1–D1, the per-slice view of Fig. 1. Values are in Table 4.

## C.3 What happens when the audit result is actually deployed

Lower regret does not by itself mean a team should act on the audit, because acting means replacing whatever the log currently recommends, and that change can be harmful. We therefore compare three arms at matched total budget, deploying the audited candidate ungated and gating it on a held-out

Table 4: **Slice contributions to pooled-policy regret** at a $35\%$ budget under the selective-router log. Each entry evaluates the *pooled* deployed policy on that slice's scoring requests only, against the pooled target policy, so an entry can be slightly negative when a slice's sub-population within a mixed context prefers a different model. The last column is the unweighted average over slices. It is a different estimator from the pooled regret of Table 3 and the two should not be compared numerically.

| Method | SimpleQA | MMLU-Pro | HLE | LiveCodeBench | ArenaHard | Unwtd. avg. |
|---|---|---|---|---|---|---|
| *A1–D0* | | | | | | |
| Trust log | 0.0711 | 0.0239 | 0.0359 | 0.0535 | 0.1311 | 0.0631 |
| Uniform audit | 0.0528 | 0.0112 | 0.0249 | 0.0624 | 0.0815 | 0.0466 |
| Raw margin, obs. surface | 0.0526 | 0.0115 | 0.0259 | 0.0541 | 0.0911 | 0.0470 |
| R-EPIG, obs. surface | 0.0532 | 0.0127 | 0.0246 | 0.0523 | 0.0864 | 0.0458 |
| CASE, deconf. surface | 0.0407 | 0.0131 | 0.0223 | 0.0502 | 0.0741 | 0.0401 |
| R-EPIG, deconf. surface | 0.0430 | 0.0142 | 0.0231 | 0.0483 | 0.0751 | 0.0407 |
| *A1–D1* | | | | | | |
| Trust log | 0.0581 | 0.0217 | 0.0356 | 0.0619 | 0.1217 | 0.0598 |
| Uniform audit | 0.0502 | 0.0113 | 0.0252 | 0.0690 | 0.0840 | 0.0479 |
| Raw margin, obs. surface | 0.0494 | 0.0132 | 0.0240 | 0.0532 | 0.0903 | 0.0460 |
| R-EPIG, obs. surface | 0.0524 | 0.0129 | 0.0239 | 0.0574 | 0.0900 | 0.0473 |
| CASE, deconf. surface | 0.0375 | 0.0122 | 0.0228 | 0.0562 | 0.0817 | 0.0421 |
| R-EPIG, deconf. surface | 0.0420 | 0.0135 | 0.0244 | 0.0544 | 0.0822 | 0.0433 |

check read either by the operational evaluator or by the target channel. Both checks cut harmful switches by roughly two thirds and are indistinguishable from each other on that outcome, but only the target channel has a verdict that is right in the limit.

Table 5: **Deployment decisions at matched total budget** ($35\%$, where every row spends $13$ cells and gated rows move $4$ to verification). A *false switch* replaces the log-derived deployment policy ($\arg\max_k M_{kj}$, *not* the incumbent router) with the audited candidate when the candidate has strictly lower target value on the untouched scoring cohort, and *false retention* is the converse. *Verdict inversion* is the fraction of contested contexts on which the checking channel's *population* comparison disagrees with the target's, a systematic error no budget removes.

| | Ref. regret ↓ | False switch ↓ | False retention ↓ | Switch rate | Cells acq. / ver. | Verdict inversion |
|---|---|---|---|---|---|---|
| *A1–D1 (stable evaluator distortion)* | | | | | | |
| CASE, ungated | $0.0338_{\pm 0.0017}$ | $0.178_{\pm 0.030}$ | $0.000_{\pm 0.000}$ | 0.99 | 13 / 0 | – |
| + same-evaluator check | $0.0374_{\pm 0.0020}$ | $0.061_{\pm 0.019}$ | $0.233_{\pm 0.039}$ | 0.56 | 9 / 4 | $4.6\%$ |
| + reference-channel check | $0.0373_{\pm 0.0020}$ | $0.072_{\pm 0.021}$ | $0.233_{\pm 0.038}$ | 0.57 | 9 / 4 | $0.0\%$ |
| *A1–D2 (evaluator differs between log and audit)* | | | | | | |
| CASE, ungated | $0.0348_{\pm 0.0020}$ | $0.189_{\pm 0.034}$ | $0.000_{\pm 0.000}$ | 0.99 | 13 / 0 | – |
| + same-evaluator check | $0.0379_{\pm 0.0020}$ | $0.050_{\pm 0.016}$ | $0.244_{\pm 0.036}$ | 0.52 | 9 / 4 | $4.8\%$ |
| + reference-channel check | $0.0377_{\pm 0.0019}$ | $0.078_{\pm 0.021}$ | $0.239_{\pm 0.035}$ | 0.56 | 9 / 4 | $0.0\%$ |

### C.4 Does any of this survive a different benchmark and a different model roster?

Every number above comes from one corpus, so its intervals describe that corpus rather than routing in general. We therefore run the identical pipeline, with the same cohorts, contexts, router, distortion formula, methods and budgets, on RouterBench, which holds $36{,}497$ requests and $11$ models across five benchmark families and shares no requests, models or scoring convention with the primary corpus. Three findings replicate and two do not, and we report both outcomes.

Selection is present, since the log misranks $46.9\%_{\pm 3.3}$ of contexts against $25.6\%_{\pm 2.9}$ for the matched common-population log. Auditing repairs part of it, taking A1–D0 trust from $0.0784$ to CASE at $0.0421$, a paired $-0.0363_{\pm 0.0042}$ that is better on every partition. A cellwise-perfect repair still leaves a measurement floor of $13.8\%_{\pm 1.5}$ identification error and $0.0053$ regret. The two findings that do not replicate are corpus-specific. First, the rule tie does not, because with $11$ models the margin heuristic loses to the decision-information rule by $+0.0140_{\pm 0.0021}$ (A1–D0) and $+0.0066_{\pm 0.0027}$ (A1–D1). Second, the cost of auditing under a common-population log does not, because here CASE minus

trust is $-0.0005_{\pm 0.0037}$ rather than $+0.0107$, consistent with RouterBench's much larger log cells leaving less variance for a 30-request readout to add.

| Acquisition rule | Corpus | Observed surface | Deconfounded surface |
|---|---|---|---|
| Smallest margin | LLMRouterBench ($K$=5) | $0.0402_{\pm 0.0019}$ | $0.0338_{\pm 0.0020}$ |
| Decision-information | LLMRouterBench ($K$=5) | $0.0391_{\pm 0.0019}$ | $0.0335_{\pm 0.0019}$ |
| Smallest margin | RouterBench ($K$=11) | $0.0467_{\pm 0.0030}$ | $0.0421_{\pm 0.0036}$ |
| Decision-information | RouterBench ($K$=11) | $0.0396_{\pm 0.0023}$ | $0.0281_{\pm 0.0029}$ |

A1–D0 at a 35% budget. The surface effect is negative in all four rows and in the matching D1 rows, while the rule effect is null at $K$=5 and large at $K$=11.

### C.5 Is the un-repaired gap a budget problem or a model problem?

CASE closes between a quarter and a third of the distance to a perfect repair, and a reader is entitled to ask whether the rest would come with more readouts. To separate the two explanations we hold CASE's allocation and budget fixed and vary only the correction, adding shrinkage, a two-way additive fit and a rank-2 completion, together with ORACLE ROW, the best-fitting correction inside the same one-offset per-model family computed from the truth with no budget at all. That last arm is the family's least-squares optimum rather than its regret optimum, so it estimates the ceiling rather than bounding it. The answer is that the shortfall is a model problem. The no-budget ceiling is $0.0318_{\pm 0.0025}$, within noise of what CASE already achieves, so more readouts would buy almost nothing, while the richer families overfit the 13 purchased cells and do worse.

| Correction, at CASE's allocation and a 35% budget | A1–D0 | A1–D1 |
|---|---|---|
| Trust the log (no correction) | $0.0499_{\pm 0.0030}$ | $0.0469_{\pm 0.0025}$ |
| Action-level (deployed) | $0.0338_{\pm 0.0020}$ | $0.0338_{\pm 0.0017}$ |
| + experiment-grounded shrinkage [Athey et al., 2020] | $0.0339_{\pm 0.0020}$ | $0.0338_{\pm 0.0017}$ |
| Two-way additive (row + column) | $0.0399_{\pm 0.0019}$ | $0.0394_{\pm 0.0024}$ |
| Rank-2 completion of the purchased bias residuals | $0.0452_{\pm 0.0019}$ | $0.0426_{\pm 0.0019}$ |
| ORACLE ROW: best-fitting action-level correction, no budget | $0.0318_{\pm 0.0025}$ | $0.0337_{\pm 0.0024}$ |
| ORACLE CELL: population-operational surface | 0.0000 | $0.0030_{\pm 0.0006}$ |

30 partitions crossed with 6 seeds. Diagnostics: $68.4\%_{\pm 2.0}$ of the selection distortion is not row-constant, the within-row standard deviation is $0.101_{\pm 0.003}$, and the fitted offsets differ from the best-fitting ones by $0.079_{\pm 0.003}$. CASE minus ORACLE ROW is $+0.0020_{\pm 0.0024}$, and two one-sided tests do *not* establish equivalence at a 0.005 margin ($p = 0.11$), so we report the 95% CI $[-0.0030, +0.0070]$ rather than claiming a tie.

### C.6 When is a deployment gate worth using?

A gate trades one kind of mistake for another, so calling it a safety measure is incomplete without saying how the two mistakes compare in cost. We therefore compute, at each budget, the ratio $c^\star = \Delta\text{retention}/|\Delta\text{switch}|$ above which gating lowers expected loss, which gives a usable rule of thumb. Under A1–D1, $c^\star = 0.98$ at a 15% budget, 1.64 at 25% and 2.00 at 35%, and for the reference channel 1.03, 1.40 and 2.21. Gating also changes regret by $-0.0051$, $+0.0021$ and $+0.0036$ respectively, so it earns its cells when the audit behind it is thin and costs value when the audit is rich.

### C.7 Are the two claimed ties really ties?

A difference that fails to reach significance is not the same as a demonstrated equivalence, and the paper leans on two such claims. We therefore report the minimum detectable paired difference at 30 partitions and run two one-sided tests against a margin fixed in advance at 0.005, a tenth of the loss of trusting the log. One claim survives as a genuine equivalence and one does not. The minimum detectable difference is 0.0031 for the CASE against R-EPIG$_{\text{corr}}$ contrast, and the two one-sided tests give $p = 1 \times 10^{-4}$ (A1–D0) and $p = 8 \times 10^{-4}$ (A1–D1), so the two rules are equivalent at that margin. The same test does not establish equivalence between CASE and the action-level ceiling, which is why that comparison is reported above as an interval rather than a tie.

### C.8 Can a team tell in advance whether to audit?

The Q6 result, that readouts thinner than the log cells they replace add error where there is no imbalance to remove, is only actionable if the imbalance can be detected before the budget is spent. From the $K$ per-model seed readouts alone, which every arm buys first, we take the dispersion of $M_{kj} - R_{kj}$ standardised by the noise it would carry if the log were common-population. This separates the two regimes and yields a decision rule that beats both fixed policies. The statistic averages $1.06{\pm}0.03$ under A0 and $1.28{\pm}0.04$ under A1, with AUC $0.79$, a best in-sample threshold of $1.17$ and accuracy $75\%$. Auditing only above that threshold gives mean regret $0.0313$, against $0.0329$ for always auditing and $0.0356$ for never auditing.

### C.9 How much does the result depend on how aggressively the router selects?

Assignment validity is half the paper, and its magnitude is set entirely by how sharply the incumbent router selects, which the main runs fix at an exploration rate of $0.15$. We sweep that rate, since $\varepsilon \to 1$ is uniform routing and reproduces A0 by construction. Both the damage and the repair scale with the selection, which is what the mechanism predicts and what makes an audit worth buying only where selection is present.

| Exploration rate $\varepsilon$ | 0.05 | 0.15 | 0.30 | 0.50 |
|---|---|---|---|---|
| Log top-1 error, A1–D0 | 0.631 | 0.675 | 0.613 | 0.569 |
| Trust the log | 0.0476 | 0.0499 | 0.0412 | 0.0368 |
| CASE | 0.0292 | 0.0338 | 0.0344 | 0.0333 |
| R-EPIG, deconfounded surface | 0.0282 | 0.0335 | 0.0330 | 0.0334 |
| CASE − trust | $-0.0183{\pm}0.0041$ | $-0.0161{\pm}0.0028$ | $-0.0068{\pm}0.0044$ | $-0.0035{\pm}0.0036$ |
| Share of the trust regret removed | 39% | 32% | 17% | 10% |

A1–D0 at a $35\%$ budget over $20$ partitions crossed with $6$ seeds, with $\varepsilon = 0.15$ taken from the main runs. The repair falls monotonically as the router explores more, from $39\%$ of the trust regret at $\varepsilon = 0.05$ to $10\%$ at $\varepsilon = 0.50$. The damage is not monotone at the sharpest setting, and the reason is support rather than bias. At $\varepsilon = 0.05$ only $28.6$ of the $40$ model–context cells appear in the log at all, against $36.2$ at $\varepsilon = 0.15$ and $40.0$ at $\varepsilon = 0.50$, and a cell that never appears is excluded from the comparison, so the surviving comparison is both smaller and easier. Aggressive routing therefore converts part of the bias problem into the support problem the last row of Table 2 records, which no audit budget addresses. What this sweep does not vary is the router *family*, which remains a cost cascade throughout.

### C.10 How much does the result depend on the shape and size of the distortion?

The evaluator distortion is a modelling choice, so a result that only holds at one strength or one functional form would be an artefact of that choice. We therefore vary the strength to the pre-specified stronger level, replace the capability-by-difficulty interaction with a context-invariant model preference, and reverse the sign of the mismatch between the log and audit streams. The measurement result holds in every variant, while the mismatch result turns out not to have a stable direction.

At $\lambda = 1.0\times$ the training median margin the reversal rate rises from $18.3\%$ to $30.6\%$ and the floor from $0.0030$ to $0.0069{\pm}0.0014$, while CASE's regret is unchanged at $0.0333{\pm}0.0026$, and the same-evaluator gate's verdict-inversion rate rises to $8.3\%{\pm}1.9$. Setting $C_{kj} = C_k$, a judge that simply prefers some models everywhere and the cleanest case of Prop. 3.1(b), cancels from the differencing entirely, yet a cellwise-perfect repair still misidentifies $10.0\%{\pm}2.1$ of contexts, so the measurement result does not depend on the capability-by-difficulty interaction. With $C^i = -\frac{1}{2}C^o$ CASE reaches $0.0313{\pm}0.0021$, so attenuation slightly worsens regret and sign reversal slightly improves it, and mismatch is real in the algebra but not directionally stable.

### C.11 How much does the result depend on the experimental settings?

Three settings were chosen rather than derived, namely how many requests each purchased cell reads, how finely requests are grouped into contexts, and which policy counts as the status quo. We sweep the readout size, the number of contexts, and the definition of the baseline the gate compares against.

Every qualitative conclusion is stable, while two magnitudes are not, which is why the main text reports the identification floor as the top of a range.

With $n \in \{15, 30, 60\}$ CASE reaches $0.0365$, $0.0338$ and $0.0338$, and the verdict-inversion rate is flat at $6.0\%$, $4.6\%$ and $5.8\%$, as a systematic error must be. Across $J \in \{6, 8, 12, 16\}$ the log misranks about two thirds of contexts, the audit reduces regret, the two deconfounded rules track each other, and a cellwise-perfect repair leaves a non-zero floor, while the identification floor ranges over $10.4$ to $18.3\%$. Taking the *incumbent router* rather than the log-derived policy as the status quo makes switching almost always beneficial (ungated false switch $0.058_{\pm 0.022}$), so that comparison is uninformative, and the main text uses the log-derived baseline because the question is whether an evaluation *conclusion* supports a change.

# D Confidence intervals for the confirmatory contrasts

The $\pm$ used elsewhere is a paired standard error over the $30$ partitions, which answers one question, namely whether a different split and acquisition draw would agree, and not two others. *Partition* is a paired $t$ interval over partitions. *Request* is a nonparametric bootstrap of the scoring cohort, $2000$ replicates with common random numbers, each run's deployed policy held fixed while $\mu^{\mathrm{ref}}$ and $\pi^{\mathrm{ref}}$ are re-estimated per replicate. *Slice* is a cluster bootstrap over the five benchmark slices, $10^4$ replicates. Five clusters make it coarse and wide by construction, and it is the closest thing here to asking whether a result would survive a different suite.

| Contrast (35% budget) | Mean | Partition | Request | Slice clusters |
|---|---|---|---|---|
| A1–D0: CASE − trust | $-0.0161$ | $[-.0219, -.0103]$ | $[-.0190, -.0132]$ | $[-.0410, -.0083]$ |
| A1–D0: CASE − R-EPIG$_{\mathrm{corr}}$ | $+0.0003$ | $[-.0019, +.0025]$ | $[-.0007, +.0013]$ | $[-.0017, +.0007]$ |
| A1–D1: CASE − trust | $-0.0130$ | $[-.0179, -.0081]$ | $[-.0159, -.0103]$ | $[-.0293, -.0087]$ |
| A1–D1: CASE − pop.-operational oracle | $+0.0309$ | $[+.0271, +.0346]$ | $[+.0278, +.0338]$ | $[+.0193, +.0444]$ |
| A1–D1: measurement floor $G_{\mathrm{meas}}$ | $+0.0030$ | $[+.0017, +.0043]$ | $[+.0011, +.0047]$ | $[-.\mathbf{0002}, +.\mathbf{0259}]$ |
| A0–D0: CASE − trust (cost of auditing) | $+0.0107$ | $[+.0067, +.0147]$ | $[+.0083, +.0132]$ | $[-.\mathbf{0013}, +.\mathbf{0142}]$ |
| A1–D1: false switch, same − ref. channel | $-0.0111$ | $[-.0269, +.0047]$ | — | — |
| A1–D2: false switch, same − ref. channel | $-0.0278$ | $[-.0514, -.0042]$ | — | — |

*Leading zeros are dropped inside the intervals.*

Three conclusions are unaffected by the scheme. The audit benefit under both evaluator conditions excludes zero everywhere, slice resampling included, so it is not carried by one benchmark. The CASE against R-EPIG tie is tight under all three, with a widest interval of $[-.0017, +.0007]$. And CASE's distance from the population-operational oracle stays large and positive. Two conclusions are weaker than the partition interval alone suggests, and the main text flags both. The measurement floor *in value* and the cost of auditing under A0 each lose separation from zero under the five-slice cluster bootstrap. Neither is surprising, given small quantities and five clusters, but neither should be quoted as established across benchmark suites. The identification-side floor does not depend on this.